

(ISTS) and intra-state transmission system (InSTS).

This persistent shortfall has led to a large backlog of projects. Combined with new generation and rising demand, the annual requirement has increased sharply. For FY2026-27, the target for the ISTS component alone has risen to about 25,146 ckm. The government's own estimate pegs the total additional ISTS requirement at 61,411 ckm by the end of FY2029-30. In addition, there will be InSTS requirements, though these are harder to assess as India is only beginning to mandate transmission adequacy at the state level.

According to industry players, there is a need for long-term transmission line planning, keeping in mind the massive solar deployment in the pipeline. Sinha explains that curtailment is happening in a few places, and while its is difficult to estimate how much, it would range between 20-40% plant-wise in various states. "We continue to work with transmission companies and the central transmission utility to ensure that the curtailment is reduced, and we can evacuate all the power. In cases where we have the general network access (GNA), and curtailment takes place, there, of course, we are reimbursed the cost on the supply basis," Sinha says. The Central Electricity Regulatory Commission has a graded financial compensation framework for power curtailment under the GNA regime which allows power producers to access the interstate transmission network.

Grid planning in India will require a range of long-term reforms to prevent build-up of large connectivity queues, growing mismatch between generation and transmission infrastructure, and the associated rise in renewable



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CMD, TATA POWER

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energy curtailment. Vineet Mittal, Chairman, Avvada, says India needs to do more aggressive grid planning.

"We should not only be planning for 2030 but should already be preparing for 2035 and beyond, keeping in mind our massive energy transition plans. Renewable-energy expansion is increasingly constrained by transmission infrastructure. Transmission planning and execution for the renewable-energy corridor should ideally be more closely integrated with the renewable-energy ministry itself to accelerate the projects," he explains.

Transmission planning is currently under the power ministry.

The growth of solar capacity across ISTS pooling stations has been rapid due to the modular nature of solar projects and their



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CHAIRMAN, AVAADA GROUP

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relatively short construction timelines. However, this has also resulted in substantial transmission infrastructure being developed for evacuation from these regions, much of which remains under-utilised outside solar generation hours. About 71% of interstate transmission corridors operate below 30% utilisation, meaning large transmission investments remain underutilised during non-solar hours, according to an IEEFA and JMK report.

Slow progress on evacuation infrastructure means renewable producers are holding back from bringing new capacity online, ultimately risking the country missing its 2030 target. India's renewable energy efficiency now depends on infrastructure upgrades. **BT**

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